

ASSIGNMENT 3

ECHO SERVER USING TCP

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Aim: To Develop a socket program to simulate Echo Server.

Echo Server : The client sends data to server. The server in turn sends the message back to the client. Send multiple lines of text.

Sample Input Output

Client Side

Client: "hi this is from ssn.

-----"

Message from Server: "hi this is from ssn.

-----"

Server Side

Message from Client: "hi this is from ssn.

-----"

Message sent: "hi this is from ssn.

-----"

Code:

Server

```
#include <stdio.h>
#include <sys/types.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

int main(int argc, char **argv)
{
    socklen_t len;
    int sockfd, newfd, n;
    struct sockaddr_in servaddr, cliaddr;
    char buff[1024];

    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    if (sockfd < 0)
    {
        perror("cannot create socket");
        return 1;
    }

    bzero(&servaddr, sizeof(servaddr));
    servaddr.sin_family = AF_INET;
    servaddr.sin_addr.s_addr = INADDR_ANY;
    servaddr.sin_port = htons(7777);
```

```

if (bind(sockfd, (struct sockaddr*)&servaddr, sizeof(servaddr)) < 0)
{
    perror("Bind error");
    return 1;
}

listen(sockfd, 2);
len = sizeof(cliaddr);

// Receiving messages
while (1)
{
    newfd = accept(sockfd, (struct sockaddr*)&cliaddr, &len);
    if (newfd < 0)
    {
        perror("Accept error");
        return 1;
    }

    while (1)
    {
        n = read(newfd, buff, sizeof(buff) - 1);
        if (n < 0)
        {
            perror("Read error");
            close(newfd);
            return 1;
        }
        else if (n == 0)
        {
            // Connection closed by client
            break;
        }

        buff[n] = '\0'; // Null-terminate the string
        printf("Received Message: %s\n", buff);

        if (strcmp(buff, "END") == 0)
        {
            close(newfd);
            break;
        }
    }
    if (strcmp(buff, "END") == 0)
    {
        break;
    }
}

close(sockfd);
return 0;
}

```

Client

```
#include <stdio.h>
```

```

#include <sys/types.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

int main(int argc, char **argv)
{
    int sockfd, n;
    struct sockaddr_in servaddr;
    char buff[1024];

    if (argc < 2)
    {
        fprintf(stderr, "Usage: %s <server-ip-address>\n", argv[0]);
        return 1;
    }

    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    if (sockfd < 0)
    {
        perror("cannot create socket");
        return 1;
    }

    bzero(&servaddr, sizeof(servaddr));
    servaddr.sin_family = AF_INET;
    servaddr.sin_addr.s_addr = inet_addr(argv[1]);
    servaddr.sin_port = htons(7777);

    if (connect(sockfd, (struct sockaddr*)&servaddr, sizeof(servaddr)) < 0)
    {
        perror("Connect error");
        close(sockfd);
        return 1;
    }

    // Sending the message
    while (1)
    {
        printf("Enter the message: ");
        if (fgets(buff, sizeof(buff), stdin) == NULL)
        {
            perror("Input error");
            close(sockfd);
            return 1;
        }
        buff[strcspn(buff, "\n")] = '\0'; // Remove newline

        n = write(sockfd, buff, strlen(buff) + 1);
        if (n < 0)
        {
            perror("Write error");
            close(sockfd);
        }
    }
}

```

```

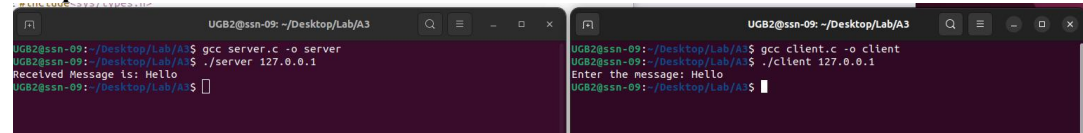
        return 1;
    }

    if (strcmp(buff, "END") == 0)
        break;
    }

    close(sockfd);
    return 0;
}

```

Output:



```

UGB2@ssn-09: ~/Desktop/Lab/A3
UGB2@ssn-09: ~/Desktop/Lab/A3$ gcc server.c -o server
UGB2@ssn-09: ~/Desktop/Lab/A3$ ./server 127.0.0.1
Received Message is: Hello
UGB2@ssn-09: ~/Desktop/Lab/A3$

UGB2@ssn-09: ~/Desktop/Lab/A3$ gcc client.c -o client
UGB2@ssn-09: ~/Desktop/Lab/A3$ ./client 127.0.0.1
Enter the message: Hello
UGB2@ssn-09: ~/Desktop/Lab/A3$

```

Server Output:

```

● nivethadhanakoti@Nivethas-MacBook-Pro A3 % gcc server.c -o server
● nivethadhanakoti@Nivethas-MacBook-Pro A3 % ./server 127.0.0.1
Received Message: Hi
Received Message: Hello
Received Message: Bye
Received Message: END

```

Client Output:

```

nivethadhanakoti@Nivethas-MacBook-Pro A3 % gcc client.c -o client
nivethadhanakoti@Nivethas-MacBook-Pro A3 % ./client 127.0.0.1
Enter the message: Hi
Enter the message: Hello
Enter the message: Bye
Enter the message: END

```

Learning Outcomes:

I have learnt to be proficient in developing socket program through echo server.